Rini Banerjee

Curriculum Vitae

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CONTACT

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EMPLOYMENT

Research Scientist Sep 2026 – present
Rigorous Engineering Ltd

Research Intern Jun 2021 – Sep 2021
Huawei Technologies Research & Development (UK) Ltd

Undergraduate Research Intern Jul 2020 – Sep 2020
Imperial College London

Undergraduate Research Intern Jul 2019 – Aug 2019
Imperial College London

EDUCATION

PhD in Computer Science Oct 2022 - present
University of Cambridge

- Thesis: Executable Separation-Logic Specifications
- Supervisors: Prof Peter Sewell & Prof Neel Krishnaswami

MEng Mathematics & Computer Science Oct 2018 - Jun 2022
Imperial College London

- First-class honours
- Master's thesis: Software Transactional Memory for Non-Volatile Memory. Supervisor: Dr Azalea Raad.
- Awards: ARM Project Prize (for thesis), Dean's List (Year 3), Corporate Partnership Programme Prize, IBM Group Project Prize

GRANTS & FUNDING

Safeguarded AI Jul 2026
ARIA

- Verified pKVM: AI-Enabled Verification of a Production Hypervisor. £3.28m.
  Peter Sewell (PI), Sean Holden (PI), Neel Krishnaswami (PI), Christopher Pulte (PI), Mantas Bakšys (co-author), **Rini Banerjee** (co-author), David Kaloper-Meršinjak (co-author), Hiroyuki Katsura (co-author), Kayvan Memarian (co-author).

Amazon Research Award Mar 2026
Amazon

- Gradual Lightweight Methods for High-Assurance Cloud Infrastructure. Automated Reasoning, Fall 2025. $79,996.00 cash award plus $10,000 in AWS credits.
  Peter Sewell (PI), Ben Simner (co-author), Kayvan Memarian (co-author), **Rini Banerjee** (co-author), Hiroyuki Katsura (co-author).

PhD Completion Grant Jun 2026
Department of Computer Science & Technology, University of Cambridge

- Awarded £5195 from Department during fourth year of PhD

PhD Completion Grant Jun 2026
Newnham College

- Awarded £1000 from College during fourth year of PhD

Google DeepMind Scholarship Apr 2022
Cambridge Trust

- Awarded on entry to PhD program, initially for 3 years, and then awarded a further 1 year of extension funding
- Total award over 4 years: £70,112

PUBLICATIONS

1. **Code-Specify-Test-Debug-Prove: Flexibly Integrating Separation Logic Specification into Conventional Workflows.** Zain K Aamer*, Rini Banerjee*, Hiroyuki Katsura*, David Kaloper-Meršinjak*, Dimitrios J. Economou, Kayvan Memarian, Dhruv Makwana, Neel Krishnaswami, Benjamin C Pierce, Christopher Pulte, Peter Sewell. PLDI 2026. [pdf] **These authors contributed equally.*
2. **Fulminate: Testing CN Separation-Logic Specifications in C.** Rini Banerjee, Kayvan Memarian, Dhruv Makwana, Christopher Pulte, Neel Krishnaswami, Peter Sewell. POPL 2025. [pdf]
3. **Smartphone Security Behavioral Scale: A New Psychometric Measurement for Smartphone Security.** Hsiao-Ying Huang, Soteris Demetriou, Rini Banerjee, Güliz Seray Tuncay, Carl A Gunter, Masooda Bashir. Preprint, 2020. [pdf]

THESES

1. **Software Transactional Memory for Non-Volatile Memory.** Rini Banerjee. MEng thesis, Imperial College London. *Awarded the ARM Project Prize.* [pdf]

TEACHING

Lecturer Easter 2026
University of Cambridge

- Course: Hoare Logic & Model Checking
- Lectured to Part II (third-year) undergraduates for Hoare Logic half of course. Set and marked exam question

Supervisor Michaelmas 2023, 2024
University of Cambridge

- Course: Semantics of Programming Languages

Undergraduate Teaching Assistant Oct 2020 - Apr 2022
Imperial College London

- Courses: Logic, Reasoning About Programs, Graphs & Algorithms, Discrete Mathematics

SUPERVISION

Muhammad Saljuk Gondal Oct 2025 - Jun 2026
MPhil thesis: Evaluating a Lua runtime for CN separation-logic testing in Fulminate. [pdf]

TALKS

Gradual Assurance for Systems Software: A Separation-Logic Approach 8 Jun 2026
Invited talk at the Iris Workshop 2026. [slides] *ISTA, Vienna, Austria*

Gradual Assurance for Systems Software: A Separation-Logic Approach 14 Apr 2026
Invited talk at the LFCS seminar. [slides] *University of Edinburgh*

Gradually Retrofitting Assurance into Systems Software: A Separation-Logic Approach 12 Jan 2026
Presented at the Theory and Practice of Static Analysis workshop at POPL 2026. [slides] *Rennes, France*

- Co-presented with Hiroyuki Katsura

Separation Logic Testing and Verification for Systems Software 7 May 2025
Invited talk at Amazon's Code Reasoning Spotlight series. [slides] *Amazon (remote)*

- Co-presented with Christopher Pulte

Fulminate: Testing CN Separation-Logic Specifications in C 14 Feb 2025
Invited talk at the Birmingham theory seminar. [slides] *University of Birmingham*

Fulminate: Testing CN Separation-Logic Specifications in C 22 Jan 2025
Presented at POPL 2025. [slides] *Denver, CO, USA*

Adventures in Specification-Based Testing 8 Oct 2024
Presented at the Big Specification: Specification, Proof, and Testing at Scale workshop. [slides] *Isaac Newton Institute, Cambridge*

- Co-presented with Benjamin C. Pierce and John Hughes

PROFESSIONAL SERVICES

Co-organiser, SANDWICH Seminar Oct 2025 - present
Department of Computer Science & Technology, University of Cambridge

Reviewer Sep - Nov 2024
POPL 2024 Artifact Evaluation Committee

EXTRACURRICULAR

Woman@CL Representative Oct 2022 - Jun 2023
Women@CL, University of Cambridge

Secretary 2021 - 2022
Imperial College Women & Non-Binary Individuals in Computing

JMC Year 2 Academic Representative Oct 2019 - Jun 2020
Imperial College Union